RTX 5090 Low-Precision Product-Pattern Width Report

Direct PTX `M - M + epsilon` product probes

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Headline finding. The earlier A-only pattern understated FP8 E4M3 range. With independent `A*B` products, FP8 E4M3 x E4M3 reaches an extreme test width of **36 bits**. That extreme case does not survive, and product-exponent scanning finds **27 effective bits** for all tested FP8 variants. FP6 and FP4 remain format-limited in this product-pattern probe.

GPU	NVIDIA GeForce RTX 5090, compute capability 12.0
Build target	`compute_120a` / `sm_120a`
Source	`src/tc_test_numerics-5090-lowp-reduction-pattern.cu`
Result files	`5090/result-5090-fp8-reduction-pattern.txt`, `5090/result-5090-fp6-reduction-pattern.txt`, `5090/result-5090-fp4-reduction-pattern.txt`

1 Scope

This report covers the standalone product-pattern probe listed on the title page. It is separate from the accumulator-boundary and repeated-MMA probes. The test uses direct inline PTX `mma.sync.aligned` instructions and does not call cuBLASLt.

The probe reports externally visible behavior for selected product positions. It does not claim to reconstruct the physical tensor-core reduction tree.

2 Method

The test maps independent product terms for output register `D0`. The selected terms are:

Role	Position	Weight in D0
`+M`	`A0.byte0 * B0.byte0`	4
`-M`	`A0.byte1 * B0.byte1`	4
`epsilon`	`A0.byte2 * B0.byte2`	4

The cross terms among those positions are checked to be absent for `D0`, so the pattern isolates three product contributions:

$$D0 = 4M - 4M + 4\epsilon$$

The test enumerates every positive finite raw value in the A format and every positive finite raw value in the B format. It forms all positive products, groups them by product exponent, and scans exponent gaps:

$$effective\ width = exponent(M) - exponent(\epsilon) + 1$$

A case counts as retained when:

```
observed > 0
abs(observed - expected) <= 0.5 * abs(expected)
```

The 50% tolerance is intentional. This pattern detects whether the small term is still visible after cancellation. It does not require exact arithmetic on the retained term, so borderline FP8 cases are reported explicitly.

3 Results

Family	Variant	Extreme width	Extreme survived	Max effective width	Format-limited
FP8	E4M3 x E4M3	36	no	27	no
FP8	E5M2 x E5M2	64	no	27	no
FP8	E4M3 x E5M2	50	no	27	no
FP8	E5M2 x E4M3	50	no	27	no
FP6	E2M3 x E2M3	12	yes	12	yes
FP6	E3M2 x E3M2	18	yes	18	yes
FP4	E2M1 x E2M1	8	yes	8	yes

3.1 Product Range

Variant	Tested pairs	Survived	Max product	Min product
FP8 E4M3 x E4M3	666	612	200704	3.8147e-06
FP8 E5M2 x E5M2	2080	1332	3.2883e+09	2.3283e-10
FP8 E4M3 x E5M2	1275	976	25690112	2.9802e-08
FP8 E5M2 x E4M3	1275	976	25690112	2.9802e-08
FP6 E2M3 x E2M3	78	78	56.25	0.015625
FP6 E3M2 x E3M2	171	171	784	0.00390625
FP4 E2M1 x E2M1	36	36	36	0.25

3.2 Extreme Cases

Variant	Width	Observed	Expected	Survived
FP8 E4M3 x E4M3	36	0	1.5259e-05	no
FP8 E5M2 x E5M2	64	0	9.3132e-10	no
FP8 E4M3 x E5M2	50	0	1.1921e-07	no
FP8 E5M2 x E4M3	50	0	1.1921e-07	no
FP6 E2M3 x E2M3	12	0.0625	0.0625	yes
FP6 E3M2 x E3M2	18	0.015625	0.015625	yes
FP4 E2M1 x E2M1	8	1	1	yes

3.3 Max-Width Retained Cases

Variant	Width	M exp	Eps exp	Observed	Expected
FP8 E4M3 x E4M3	27	17	-9	0.0078125	0.0153809
FP8 E5M2 x E5M2	27	31	5	128	240
FP8 E4M3 x E5M2	27	24	-2	1	1.96875
FP8 E5M2 x E4M3	27	24	-2	1	1.96875
FP6 E2M3 x E2M3	12	5	-6	0.0625	0.0625
FP6 E3M2 x E3M2	18	9	-8	0.015625	0.015625
FP4 E2M1 x E2M1	8	5	-2	1	1

The FP8 max-width retained cases are edge cases. The observed value is roughly half of the expected epsilon contribution, but remains positive and within the retention tolerance. Therefore, **27 bits** means the widest detectable retained epsilon under this criterion, not exact 27-bit arithmetic.

4 Format Analysis

4.1 FP8 E4M3 x E4M3

The earlier A-only pattern understated the available range. With A and B both participating:

```
max product = 200704
min product = 3.81469727e-06
product exponent gap = 17 - (-18) + 1 = 36
```

The extreme case is lost. The widest retained case is 27 bits, with M at exponent 17 and ϵ at exponent -9. This is not format-limited.

4.2 FP8 E5M2 x E5M2

E5M2 has a wider exponent range than E4M3, so the extreme product width reaches 64 bits. The extreme case is lost, and the widest retained case is again 27 bits. More representable product range does not increase the observed retained width for this pattern.

4.3 FP8 Mixed E4M3/E5M2

Both mixed variants reach an extreme width of 50 bits and a max effective width of 27 bits. The two directions match in this selected layout, but that should not be generalized to every possible operand placement or output lane without additional layout sweeps.

4.4 FP6 E2M3 x E2M3

The entire tested product range is 12 bits, and the extreme case survives exactly. This result is format-limited: it proves the tested path preserves the full E2M3 product range used here, but it does not prove the internal retention width is only 12 bits.

4.5 FP6 E3M2 x E3M2

The tested range expands to 18 bits because E3M2 has more exponent range than E2M3. The extreme case survives exactly, so this is also format-limited. The test does not find a failing FP6 E3M2 product-pattern case.

4.6 FP4 E2M1 x E2M1

The tested product range is 8 bits. The extreme case survives exactly. This is strong evidence that the tested path can retain every product gap expressible by FP4 E2M1 in this layout, but it cannot bound a wider internal tree because the format cannot generate a wider product gap.

5 Edge Cases and Limitations

- The result is layout-sensitive. Other positions, lanes, or output registers may have different association behavior.
- FP8's 27-bit retained cases pass because epsilon remains visible and within the 50% retention tolerance, not because the result equals the expected value.
- FP6 and FP4 are format-limited in this test. Their max effective width is the largest width generated by the product format here.
- The grouping key is product exponent. Products in the same exponent bucket can have different significands.
- This test covers direct `mma.sync.aligned.m16n8k32` PTX and one selected output register. It does not cover block-scaled MXFP instructions or full GEMM accuracy.

6 Verification

```
make -B test-5090-fp8-reduction-pattern \
        test-5090-fp6-reduction-pattern \
        test-5090-fp4-reduction-pattern

python3 run_tests.py -o 5090 \
        5090-fp8-reduction-pattern \
        5090-fp6-reduction-pattern \
        5090-fp4-reduction-pattern

compute-sanitizer --tool memcheck ./test-5090-fp8-reduction-pattern
compute-sanitizer --tool memcheck ./test-5090-fp6-reduction-pattern
compute-sanitizer --tool memcheck ./test-5090-fp4-reduction-pattern
```

The generated SASS contains direct `QMMA.16832` instructions:

Target	Confirmed instruction family
`test-5090-fp8-reduction-pattern`	`QMMA.16832.F32.E4M3/E5M2`
`test-5090-fp6-reduction-pattern`	`QMMA.16832.F32.E2M3/E3M2`
`test-5090-fp4-reduction-pattern`	`QMMA.16832.F32.E2M1`